Saturation-aware certification of active chatter control in thin-walled milling: implementation-exact admissible sets, causal saturation islands, and the collapse of the linear controller ranking

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Abstract

Regenerative chatter, not machine capability, caps the material removal rate of thin-walled aerospace parts. Active vibration control can raise it, yet industrial uptake remains marginal: mainstream designs assume unbounded linear actuation and a time-invariant workpiece, whereas shop-floor piezoelectric actuators saturate at productive depths of cut and cutting-point dynamics vary with tool position and removal. This paper develops a saturation-aware, position-scheduled strategy treating both nonidealities as design objects: a gain-scheduled H-infinity controller, synthesized over the NC-known tool-path and removal schedule, is certified regionally on the saturated periodic delayed loop by maximal saturation-free admissible sets on a sampled-data semi-discretization lifting — implementation-exact (control rate, computation delay, amplifier bound), matching a nonlinear simulator's clip onset within 2% at reference points — and swept into a permissible depth-of-cut envelope under a declared surface-defect tolerance. On a benchmark validated against published measurements, certification overturns the linear ranking: the scheduled design's 3.2x linear worst-position advantage over the best fixed-gain design collapses to 1.3x at 1 μm tolerance and inverts to 0.79x at 20 μm, while at the hardest position it cuts vibration by 58% at 49% of its RMS voltage, clipping fifty-fold less — converting saturation from a hidden failure mode into a certified planning constraint.

Keywords: active vibration control, milling chatter, actuator saturation, maximal output admissible set, saturation island, gain scheduling

1. Introduction

The active-control literature for thin-wall milling has reached methodological maturity on an idealized plant: a linear actuator of effectively unbounded authority acting on a workpiece frozen at one nominal configuration. Within that frame, optimal, robust and μ-synthesis designs [1, 2, 3] and adaptive or learning controllers [4, 5] demonstrate substantial — in the best cases order-of-magnitude [6] — gains in stable depth of cut, in simulation or in laboratory cuts conducted inside the actuator's linear range. Two simplifications underwrite the mainstream synthesis lineage. First, actuator limits are absent from the design model: the voltage bound of a piezoelectric patch appears, if at all, as an a-posteriori check; the exceptions, discussed below, either avoid the constraint or average the dynamics. Second, the position- and removal-induced variation of the modal participation at the cutting point is ignored, wrapped into norm-bounded uncertainty at the price of conservatism [1], or tracked heuristically and reactively — position-varying PD gains [7], adaptive re-tuning [4], LPV scheduling on quasi-static setup parameters [8] — without a stability certificate for the delayed periodic loop.

Neither simplification survives contact with production economics, because both bind exactly where productivity is decided. Saturation binds first at high depth of cut: recent measurements document "saturation islands" below the linear stability boundary in which the actuator clips — and in some of which the resulting nonlinearity triggers chatter below the linearly predicted boundary [9] — so a linearly certified envelope overstates the safe region precisely where the removal rate is highest — an unacceptable failure mode for unattended machining. Existing responses either avoid the constraint through gain maps optimized to remain unsaturated [6], or establish restricted-initial-set invariance for bespoke adaptive laws on Fourier-averaged dynamics [10]; the requisite saturated-delay certificate machinery — generalized-sector conditions and maximized domains of attraction for delayed saturated systems [11, 12], integral quadratic constraints for time-periodic delayed feedback [13] — exists off the shelf and has never been applied to machining. What no published work provides is what a process planner needs: a computable, maximized region of initial conditions and cutting parameters over which a *given* controller of standard architecture, behind a *given* amplifier, is certified stable on the unaveraged time-periodic delayed dynamics. Spatial dependency binds second: the authority of any fixed design collapses where the dominant mode's participation at the tool position weakens, so the worst position along the path — not the average — caps the feed-through depth of the entire pass [14]. The two limits compound: where authority drops, the controller demands more voltage, and saturation arrives sooner.

This paper closes that gap with a control strategy in which the actuator bound and the tool-path variation are first-class design objects. Its contributions are three. (i) Building on the position- and removal-scheduled H_∞ synthesis of [14], the actuator bound is brought inside the scheduled design flow as a certified constraint: every operating point handed to process planning carries an implementation-exact certificate of saturation-free operation with a quantified physical perturbation tolerance. (ii) Stability of the saturated, time-periodic, delayed closed loop is certified regionally, for a given fixed controller of standard architecture, on the unaveraged periodic dynamics: a sampled-data semi-discretization lifting (finite control rate, computation delay, DAC-level deadzone) reduces the loop — formulated on the variational dynamics about the verified-unsaturated forced periodic response, with the phase-resolved voltage headroom as the deadzone bound — to a finite-dimensional periodic system whose maximal saturation-free admissible set (the Gilbert–Tan construction [15], extended to the periodic voltage constraint) is computed exactly, solver-free, and is exact in the reported perturbation direction — validated to 0.01–1.8% against an independent nonlinear simulator's measured clip onset (Section 3). (iii) The certificates are swept into a certified permissible-envelope (Section 4) that turns the measured island phenomenon into an a-priori certified planning constraint; across the operating grid, under identical certificate machinery, certification is shown to *overturn* the linear ranking of scheduled versus fixed-gain designs as the declared tolerance grows — to the authors' knowledge the first quantified demonstration, simulation-based on a measurement-anchored benchmark, that linear stability margins can misrank milling controllers under saturation. Two saturation islands are then demonstrated with *causal* attribution (Section 5), and the saturated regime beyond the clip boundary is analysed with circle-criterion machinery including a structural impossibility result and a certified anti-windup negative result (Section 6). All of this is developed and quantified on the thin-walled plate benchmark; as an independent external cross-check, Appendix A reproduces the published saturation-island measurements of [9] on their own published parameters, confirming that the same certificate machinery predicts a measured island pattern on a different rig.

2. Benchmark, models, and controllers

The benchmark is the thin-walled plate rig of [1], modeled and validated in the companion study [14]: a Mindlin plate finite-element model with a surface-bonded piezoelectric patch (induced-moment coupling), modal reduction (5-mode design model, 12-mode evaluation model), helical multi-tooth regenerative milling force, and an eddy-current displacement sensor. The model chain is anchored to the published measurements of the rig: natural frequencies within 1.6–3.8%, FRF-level match of receptance and voltage-to-displacement chains including antiresonance placement, and reproduction of the measured open-loop stability pattern under the published force calibration [14]. The production speed used throughout is 4.9 krpm (3-tooth cutter); tool position x_T ranges over the 100 mm plate edge.

Two controllers from the same design family are compared under identical certificate machinery: the position/removal-scheduled LPV H_∞ controller (PS-LPV) of [14] and the best frozen H_∞ design of the same family (the strongest non-scheduled baseline). Both are *deployed* controllers: actuator roll-off filter in series, controller order reduction to the real-time pole cap, and the implementation timing of the real-time target — zero-order hold at the control rate with one control period of computation delay. The amplifier saturates at $V_{\max} = \pm 150\,\mathrm{V}$; saturation acts on the commanded voltage at the DAC. Linear deployment certificates (spillover small-gain, exact sampled-loop spectral radius, closed-loop stability lobes) are inherited from [14]; this paper adds the saturation layer.

3. Implementation-exact certification of the saturated loop

3.1. Sampled-data period lifting

The closed loop is lifted per tooth-passing period τ at the control rate: $m = \lfloor \tau f_c \rceil$ ticks of length $\Delta t = \tau/m$ (at 4.9 krpm and $f_c = 50\,\mathrm{kHz}$, $m = 204$ and Δt deviates from the true control period by 0.04%, declared). The state $z = [x_p;\, x_K;\, u_{\mathrm{pend}};\, w_{1:m}]$ collects the plant modal state, the discrete controller state, the pending voltage (the computation delay made explicit), and the ring buffer of milling-point deflections that realizes the regenerative delay of exactly m ticks. Within each tick the plant evolves by its exact zero-order-hold map with the per-tick-averaged directional cutting coefficient $a_4(t)$; at each tick the controller output v_k is computed, clipped by the amplifier dead-zone $\psi_k = \mathrm{dz}_{V_{\max}}(v_k)$, and latched into the pending slot. The homogeneous period map Φ, the affine forced term (tooth-passing feed ripple), the voltage read-out rows, and the deadzone-injection columns are assembled in one pass; the deadzone thus acts exactly where the physical DAC clips, with no intra-interval hold approximation on the nonlinearity.

3.2. Forced orbit and phase-resolved headroom

The periodic forced orbit z^* and its commanded voltages v_k^* follow from the affine period map. All certificates are formulated on the variational dynamics about z^*, which is required to be strictly unsaturated; the phase-resolved headroom $\bar{u}_k = V_{\max} - |v_k^*|$ then bounds the shifted deadzone. The lifted forced voltages match the settled nonlinear simulator at the reference points (regression-tested to a 3% tolerance; the clip-onset validation of Table 1 constrains the same quantity indirectly to sub-percent accuracy at the validation points, since a headroom error maps directly into an onset error). This accuracy is load-bearing: near the envelope boundary a 1% error in the forced peak is a tenfold error in headroom. (A continuous-time lift with a Padé latency model, used for cross-validation against semi-discretization stability lobes, underestimates the forced peak by ≈5–13% as depth grows and is therefore *not* used for certification.)

3.3. Region-of-linearity certificate

Because the forced orbit is strictly unsaturated, any variational trajectory whose commanded voltage never exceeds the local headroom keeps the deadzone inactive, evolves exactly linearly, and decays whenever the Floquet radius $\rho(\Phi) < 1$. The maximal such region is the maximal output admissible set [15] of the period map under the periodic voltage constraint,

$$\mathcal{O}_\infty = \left\{ \delta z : \ |v_k^* + R_k \Phi^j \, \delta z| \le V_{\max} \ \ \forall j \ge 0, \ k = 1..m \right\}, \tag{1}$$

with R_k the voltage read-out after k ticks; $\mathcal{O}_\infty$ is invariant by the shift property since v^* is periodic. Its extent along the physical perturbation direction e_h — a step of height h left in the stored surface by the previous pass, certified for both signs — is closed-form:

$$h_{\max} = \min_{j,k} \ \frac{V_{\max} - |v_k^*|}{|R_k \Phi^j e_h|}, \tag{2}$$

computed by direct simulation of the unit step, in milliseconds and without any semidefinite solver. (Periodic generalized-sector LMI formulations of the same certificate were attempted first and are memory-infeasible at the lifted dimension on commodity hardware; Eq. (2) is not a fallback but *tighter* in the reported direction.) Declared scope: the step edge crosses at the lift's tick-0 phase (the simulator cross-check injects at the same phase); positive-branch values above the feed per tooth certify the contact-retaining model only; sensor noise is excluded; per-tick holds of a_4 and of the delayed surface are covered by the simulator validation below.

3.4. Validation against the nonlinear simulator

The ground truth is an independent 100 kHz nonlinear time-domain simulator (loss-of-contact chip nonlinearity, discrete controller, saturation, measurement chain) inherited from [14] and extended with surface-step injection; for this cross-check sensor noise and loss of contact are disabled, matching the certificate's contact-retaining scope (the island and hard-condition simulations below use their own declared switches). The onset of clipping — the smallest injected step that saturates the amplifier — is bisected to 6 nm resolution and compared with the certificate in Table 1:

Table 1: Certificate vs. measured clip onset (PS-LPV, $x_T = 50$ mm, 4.9 krpm; both step signs).

a_p	certificate (±)	model $+h/-h$	simulator $+h/-h$	deviation
0.3 mm	56.8 μm	97.68/56.81 μm	97.57/56.80 μm	0.01–0.11%
1.0 mm	6.9 μm	45.10/6.90 μm	45.07/7.03 μm	0.05–1.8%

The certificate equals the binding (negative-step) onset to 0.01–1.8% — it is tight, not merely sound, at the validation points (one controller, one position, two depths: the declared scope of this cross-check).

4. Certified envelopes and the collapse of the linear ranking

Sweeping Eq. (2) over the operating grid partitions it, per controller, into four regions: *certified* at the declared tolerance h_{req}; *linear-only* (linearly stable but certificate-refused — the exposure zone where islands can live); *forced-saturated* (the orbit itself clips: refused at any tolerance); and linearly unstable (Fig. 1a). Certified depths are prefix-connected: the reported value is the lower edge of the first refusal, so a feasible pocket above a refused band is never reported as "the" certified depth.

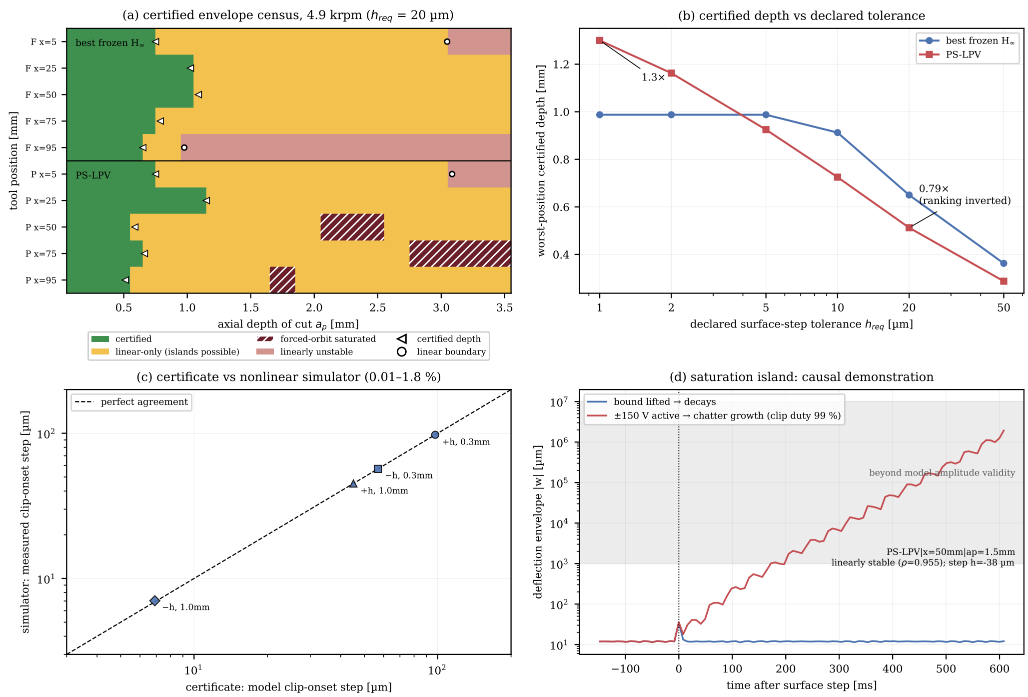

Figure 1: (a) Certified envelope census at 4.9 krpm ($h_{\mathrm{req}} = 20\,\mu$m). (b) Worst-position certified depth vs. declared tolerance: the ranking inversion. (c) Certificate vs. nonlinear-simulator clip onset (parity, both signs). (d) Saturation island, causal demonstration: the same $-38\,\mu$m step decays with the bound lifted and grows into chatter at 99% clip duty with the ±150 V bound active.

Table 2: Worst-position depths [mm] at 4.9 krpm, $x_T \in \{5, 25, 50, 75, 95\}$ mm. All *certified* worst positions bind at $x_T = 95$ mm; the PS-LPV *linear* worst binds at $x_T = 5$ mm (its linear boundary exceeds the declared 5 mm search cap at the other positions). "Linear" = sampled-loop Floquet boundary from the same lifting.

	linear	@1 μm	@2 μm	@5 μm	@10 μm	@20 μm	@50 μm
best frozen H_∞	0.98	0.99	0.99	0.99	0.91	0.650	0.36
PS-LPV	3.09	1.30	1.16	0.93	0.73	0.513	0.29
ratio	3.16×	**1.32**×	1.18×	0.94×	0.79×	**0.79**×	0.79×

Table 2 is the headline result. The scheduled design's 3.2× linear worst-position advantage collapses to at most 1.32× the moment saturation is certified — even at the tightest $1\,\mu$m tolerance — crossing below parity at $\sim 5\,\mu$m (0.94×) and settling at 0.79× from $10\,\mu$m upward: the scheduled controller's higher authority amplifies both the forced tooth-passing voltage ripple and the voltage response to surface defects, consuming headroom precisely where the linear analysis promises depth. Forced-saturated zones appear *only* for the scheduled design (Fig. 1a). At the hard condition ($x_T = 95\,$mm, $a_p = 2\,$mm, sensor noise and loss of contact active) the scheduled design still dominates operationally — 6.47 vs. $15.48\,\mu$m RMS (58% reduction) at 41.0 vs. 84.5 V RMS (49%) — but the frozen design operates 14.7% clipped versus 0.28% (≈50-fold less), and the frozen loop at that point is in fact linearly *unstable* in the implementation-exact lift ($\rho = 1.41$): its bounded response is a purely nonlinear clipped equilibrium, quantifying how misleading the linear picture is there.

5. Saturation islands: causal demonstration

An operating point exhibits a saturation island when it is linearly stable yet a finite physical perturbation triggers sustained chatter, *and* the same perturbation decays once the amplifier bound is lifted — the causal control that attributes the instability to saturation and nothing else. The protocol escalates surface steps of both signs; sign matters through the unilateral chip $h_c = \sin\varphi\,(f_t + w - w_\tau)$: a positive step thins the chip into air cutting (self-limiting), a negative step thickens it (unbounded force demand) — the negative sign is also the certificate's binding one.

At 4.9 krpm, two islands are confirmed, both for the scheduled controller at $x_T = 50\,$mm: at $a_p = 1.5\,$mm the loop ($\rho = 0.955$) tolerates $\pm 18.8\,\mu$m but a $-37.6\,\mu$m step latches into chatter growth at 99.4% clip duty; at $a_p = 2.0\,$mm the same occurs at $-50\,\mu$m (99.5%). In both cases the identical step decays with the bound lifted (Fig. 1d). The frozen design produced no island at any of the tested points up to $\pm 500\,\mu$m steps with up to 4.5% transient clipping. The certified $h_{\max}$ at the island points ($1.88\,\mu$m and $0.016\,\mu$m) sits 20× and ∼3100× below the island onsets: the certificate's refusal-to-clip conservatism, quantified against the true basin boundary.

6. The saturated regime: limits, margins, anti-windup

Beyond the clip boundary the loop is $\delta v = \delta v_{\text{free}} + L\,\psi$ with $\psi_k = \mathrm{dz}_{V_{\max}}(v_k^* + \delta v_k)$ in the componentwise sector $[0, 1]$, and $L(z)$ the period-lifted transfer from the deadzone injections to the variational voltages. Two sufficient global tests are computed on an eigenvalue-anchored frequency grid: the ℓ_2 small gain $\gamma_2 = \max_\theta \bar{\sigma}(L)$ and the sharper discrete circle-criterion level $c = \max_\theta \lambda_{\max}\big(\mathrm{Re}\, L\big)$; $c < 1$ (with linear stability and an unsaturated orbit) certifies global ℓ_2-bounded recovery from arbitrarily deep clipping.

Three findings shape the honest picture. First, a *structural impossibility*: above the open-loop stability boundary (0.253 mm at $x_T = 50$ mm) no global sector-$[0, 1]$ certificate can exist for any controller — the sector's upper edge contains the fully-clipped, i.e. open, loop — so regional certificates above that depth are a necessity, not a conservatism artifact. Second, the measured levels are dominated by the *controller*: along the $x_T = 50$ mm slice c moves only from 1.54 to 2.16 (frozen) and from 9.66 to 9.70 (PS-LPV) between near-zero cut and the island depth — a 6× larger deadzone-loop level for the island-prone high-authority design at low depth (4.5× at the island depth), qualitatively aligned with its susceptibility. The ranking is not universal: the island-free frozen point at $x_T = 95$ mm carries the largest level of all ($c = 13.4$) — the explicit counterexample showing that worst-case-phase conservatism dominates; the causal experiments of Section 5 remain the ground truth, and Zames–Falb multipliers — applicable since the shifted deadzone is monotone though non-odd — are the identified sharpening route. Third, a certified *anti-windup negative result*: back-calculation anti-windup enters the lift only through the deadzone-injection columns and provably cannot change any linear certificate; a line search on its gain at the representative low-depth point ($x_T = 50$ mm, $a_p = 0.1$ mm) returns zero improvement of c for *both* designs — within the classic anti-windup family and this certificate, no benefit is available at the audited point, and a certified anti-windup gain requires richer parameterizations or the multipliers above.

7. Discussion and limitations

The certificates delivered here refuse clipping rather than exploit it; Section 6 shows that above the open-loop boundary this is not a weakness but the only option for sector-global guarantees, and quantifies the conservatism against true basins (20–3100× at the island points). All certificates are nominal-model and deterministic: robustness to cutting-coefficient dispersion and modal tolerance is assessed statistically in the companion campaign [14]; sensor noise is excluded from the certificate and present in the evaluation simulations. The external cross-check (Appendix A) carries declared reductions (single mode, no edge coefficients, no runout, assumed feedback rate, one calibrated sign). The tolerance-resolved ranking of Table 2 is specific to this benchmark and controller family; its methodological point — that linear margins can misrank controllers under saturation, and that the certified ranking depends on the declared tolerance — is general, and we propose the tolerance-resolved certified envelope as the object that active chatter control papers should report alongside stability lobes.

8. Conclusions

(1) An implementation-exact, solver-free regional certificate for saturated active chatter control was constructed from the maximal output admissible set of a sampled-data period lifting, validated to 0.01–1.8% against an independent nonlinear simulator's measured clip onset. (2) Sweeping it yields tolerance-resolved certified envelopes under which the linear controller ranking collapses ($3.2\times \to 1.32\times$) and inverts ($0.79\times$ at $20\,\mu$m) — saturation certification changes which controller wins. (3) Two saturation islands were demonstrated with causal attribution to the amplifier bound, and the certificate quantifies its own conservatism against them. (4) In the saturated regime, a structural impossibility result scopes what any global certificate can promise above the open-loop boundary, and a certified negative result shows classic back-calculation anti-windup cannot improve the circle-criterion level for either design at the audited point. (5) As an independent external cross-check (Appendix A), the same certificate machinery reproduces the published saturation-island measurements of [9] on their own published parameters (median forced-force ratio 1.04 over the ten comparable cells; both islands present, with declared placement misses), connecting their frequency-domain saturation model to the present certificates and indicating the island mechanism to be forced-response saturation of a mode inside the tooth-passing band.

Data availability

The complete model, certificates, campaigns, and every number in this paper are reproducible from the repository accompanying the manuscript (scripts and machine-readable records referenced throughout).

Appendix A. External validation: reproduction of the published saturation islands

The plate results of Sections 3–6 rest on the thin-walled benchmark alone. This appendix adds an independent external anchor by reproducing measured saturation islands from a *different* rig on its own published parameters, with the same lifting and certificates and no plate-specific tuning.

The island measurements of [9] — a robotic-milling flexure ($f_n = 128.1$ Hz, $\zeta = 1.48\%$, $k = 1.13\times10^7$ N/m) under direct velocity feedback (253 V/(m/s)) through an 8.4 Hz proof-mass actuator (3 N/V) with a 27 N saturation threshold, cutting Al 7075-T6 with a 4-tooth Ø16 mm 45°-helix cutter at half-immersion down milling, $f_t = 50\,\mu$m — are re-simulated on the published parameters with the same lifting and certificates (single-mode reduction, untabulated edge coefficients set to zero, 10 kHz feedback rate assumed, and the regenerative sign — absent from the published regeneration-free model — calibrated once against the published uncontrolled boundary; all declared). The calibrated model reproduces the published uncontrolled boundary shape in the island band — a high left flank (model 14–18 mm at 1800–1950 rpm against the published flank rising to ∼24 mm at 1950 rpm: a declared underestimate) collapsing to ∼2 mm above 2050 rpm in both — and, as a cross-configuration consistency check only, gives 0.9 mm at 2800 rpm against the 1.2 mm the paper reports for its earlier (129.3 Hz) clamping configuration. With no further tuning (Fig. A.1):

- of the thirteen measured amplitudes printed in the paper's island figure, ten admit a forced-response comparison (the other three lie above the calibrated model's linear boundary where the experiment measured cuts — a declared discrepancy of the single-mode reduction); over the ten, the median model/measured ratio is **1.04**, eight cells within ±30% and a worst case of +55% at 1900 rpm / 5 mm, consistent with the untabulated edge coefficients and measured runout;

- *both* published islands appear as the certificate census's forced-saturated zones: the central island at its 4–8 mm rows (model 1900–1950 rpm; the published extent reaches ∼1850–2000 rpm and includes measured saturation at the 10 mm row that the model certifies saturation-free — a

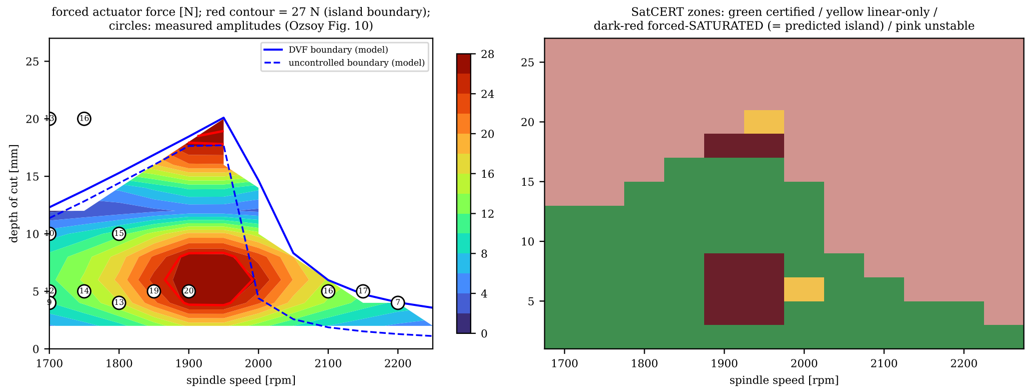

Figure A.1: Reproduction of the published saturation islands on the published parameters. Left: forced actuator-force map with the 27 N island contour and the measured amplitudes of [9] (circles). Right: certificate zones; the forced-saturated zones are the predicted islands.

declared miss), and the upper island at 18 mm against the published ∼20 mm (where the single-mode model is already linearly unstable); the saturation-free band between them, which the paper highlights at ∼15 mm, appears in the model at 10–16 mm; the paper's third band of marginal-chatter cells at 2050–2250 rpm lies along the model's stability boundary, as described there;

- a causal probe at 1900 rpm / 5 mm runs at 44.6% clip duty with mildly degraded RMS and no chatter — consistent with the paper's saturated-but-stable designations in the island interior (this is also the +55% worst cell above: the paper's measured 20 N there sits below the threshold the model crosses).

The reproduction also settles the mechanism: the published islands are *forced-response* saturation of a mode lying *inside* the tooth-passing band (128.1 Hz; the first harmonic resonates at 1921 rpm, the measured island center) — exactly the forced-saturated zones our machinery predicts, with regeneration included where the published frequency-domain model omits it. A class-representative counterfactual with the mode *below* the band (25 Hz vs. 100–200 Hz forcing) produced *no* island up to ±500 μm surface steps (≤4% transient clip duty), a step range in which the plate's scheduled H_∞ had long since latched into chatter: a clipped velocity feedback still outputs a dissipative $-\dot{x}$-signed force, whereas a clipped high-gain model-based loop loses its stabilizing phase. Island susceptibility is therefore set not by clipping depth but by the *phase integrity of the clipped feedback* and the spectral placement of the dominant mode — a design-relevant conclusion beyond certification.

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